

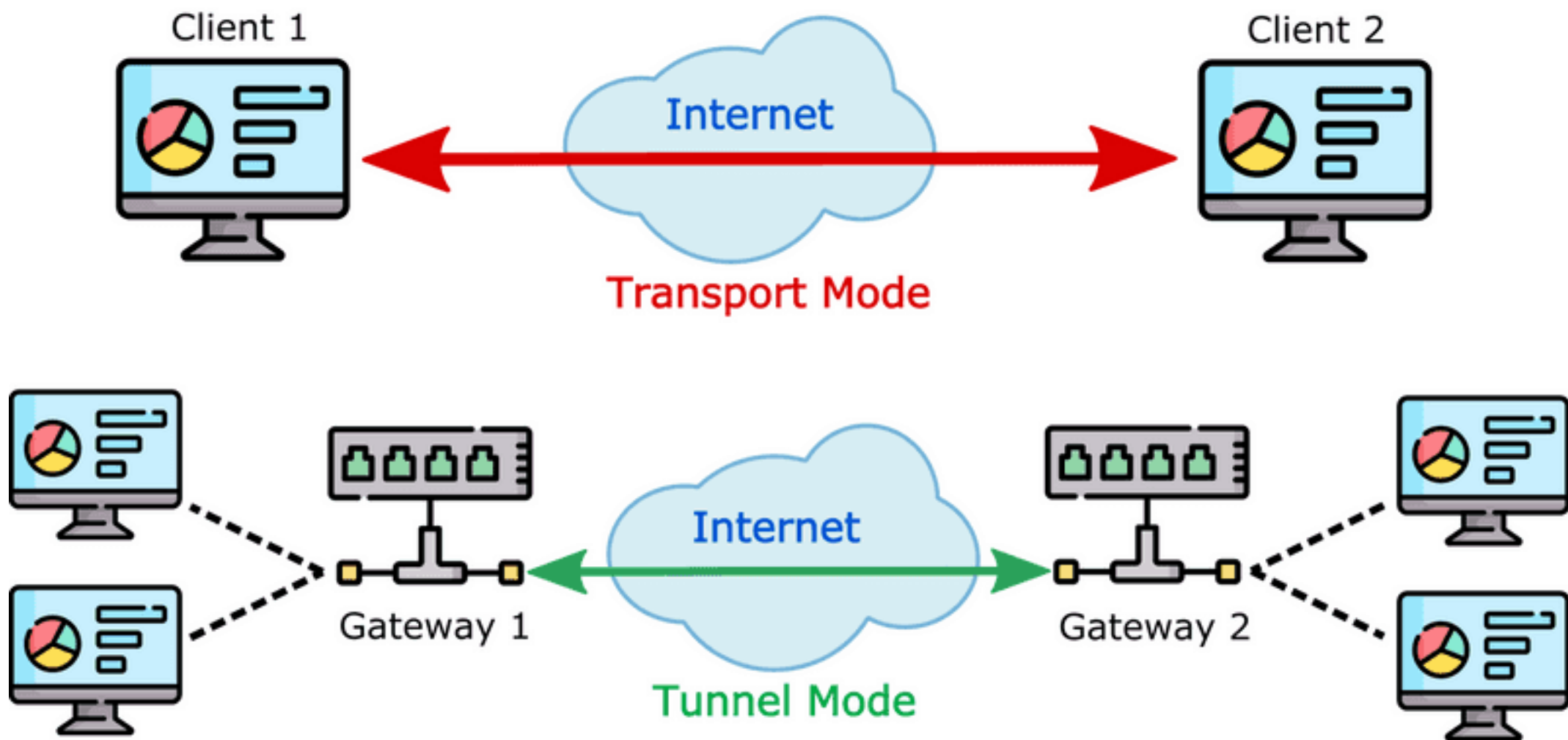
IP-Security Authentication Header

Introduction

- IPsec (Internet Protocol Security) is a suite of protocols and algorithms for securing data transmitted over the internet or any public network.
- The Internet Engineering Task Force, or IETF, developed the IPsec protocols in the mid-1990s to provide security at the IP layer through authentication and encryption of IP network packets.

- To authenticate data packets and guarantee their integrity, IPsec includes two protocols.
 - AH (Authentication Header) protocol
 - ESP (Encapsulating Security Payload) protocol.
- Both protocols, in turn, support two encapsulation modes
 - Tunnel mode
 - Transport mode
- These modes serve different purposes and provide varying levels of security.

IPSec Modes



Tunnel Mode:

- In Tunnel Mode, **the entire original IP packet is encapsulated in a new IP packet. The new packet is then encrypted and sent over the network.**
- This mode is used when two entire networks need to be connected over a public network, such as the Internet.
- In Tunnel Mode, the entire IP packet, including the original source and destination addresses, is encrypted, providing end-to-end security.

Transport Mode:

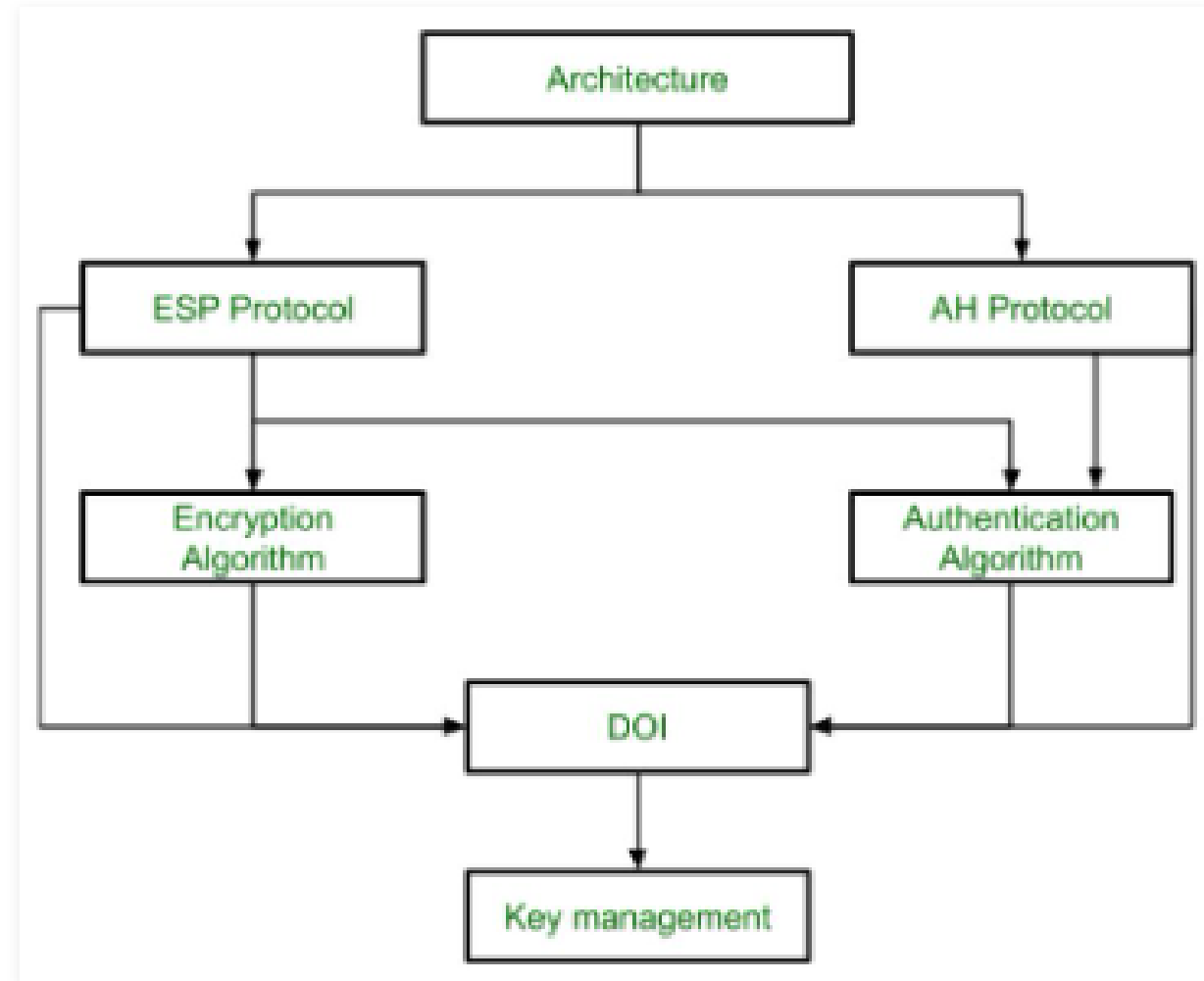
- In Transport Mode, **only the payload of the IP packet** is encrypted, not the entire packet.
- This mode is used when a single host needs to communicate with another single host over a public network.
- In Transport Mode, only the data being transmitted is encrypted, not the header information such as the source and destination addresses.

IPSec Protocol

- IPSec uses two distinct protocols: Authentication Header (AH) and Encapsulating Security Payload (ESP).
- The AH protocol provides a mechanism for **integrity and authentication** only.
- ESP can be used with **confidentiality only, authentication only, or both confidentiality and authentication**.
- AH provides data integrity by using an authentication algorithm. It does not encrypt the packet.
- ESP typically protects the packet with an encryption algorithm and provides data integrity with an authentication algorithm.

Architecture of IPSec

- DOI – Domain of Interpretation
 - It will have the identifiers or the id's of all the approved authentication and encryption algorithms



Authentication Header

Authentication Header (AH): It provides authentication and integrity (no confidentiality)

No encryption

: can provide security against replay attack using sliding window

Replay attack allows a bad guy to resend the intercepted contents between two point

Encapsulation Security Protocol (ESP): It provides authentication, integrity and confidentiality

Encryption

Transform Set

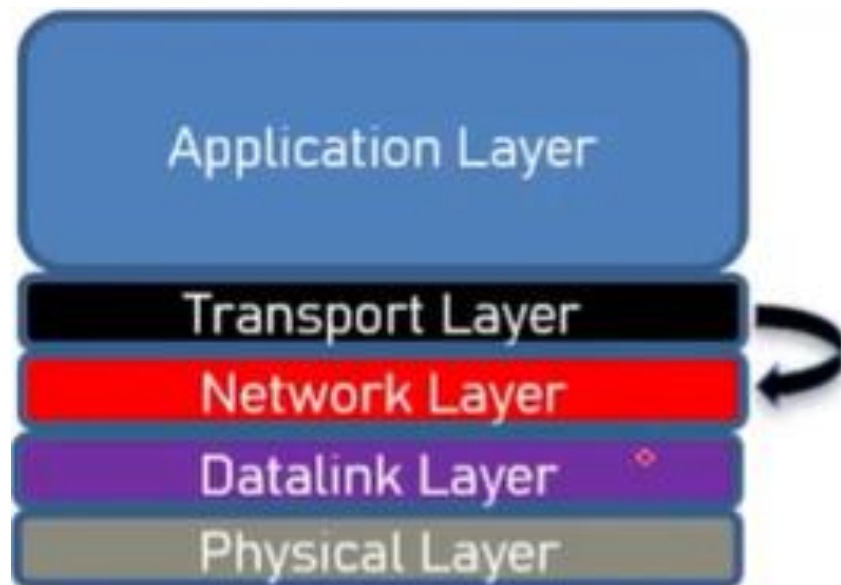
Encapsulation protocol (AH, ESP)

Encryption

Hashing

Tunnel mode (Transport or Tunnel)

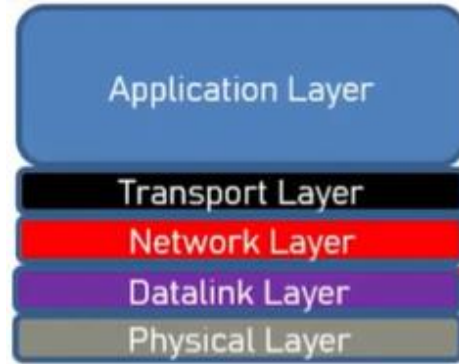
Authentication Header (AH)



IPsec provides security at layer-3 (Network Layer)



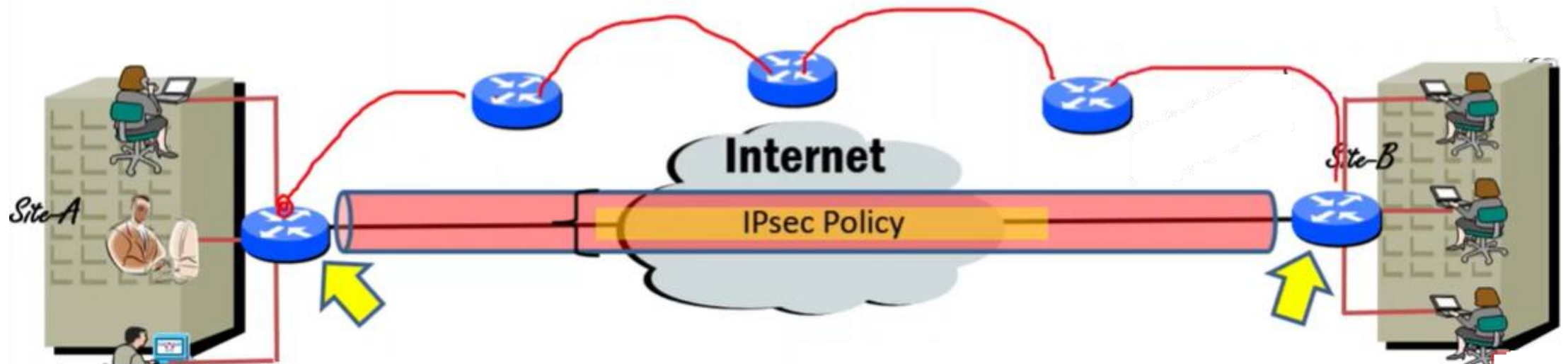
Authentication Header (AH)



Transport Mode – no protection of IP header



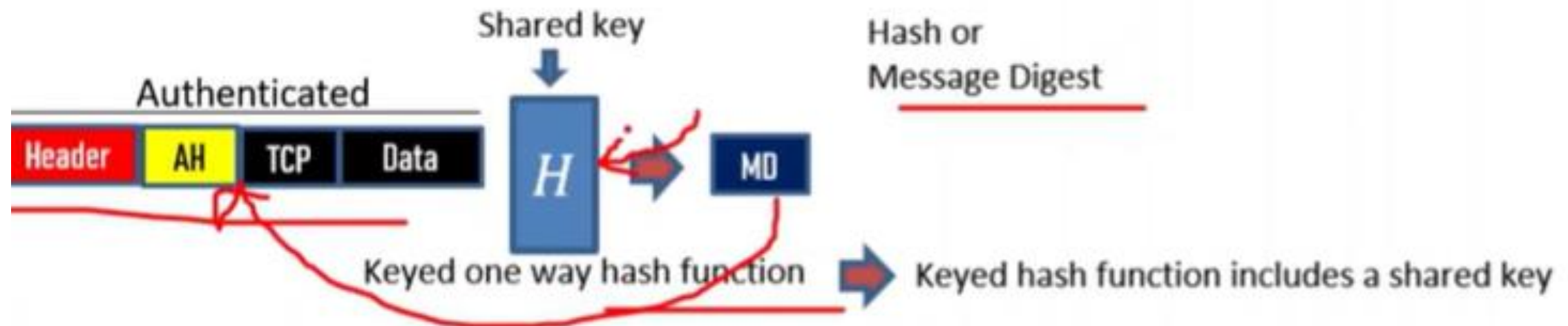
The original IP address is used to route the packet



Authentication Header (AH)

Transport Mode: No protection of IP header

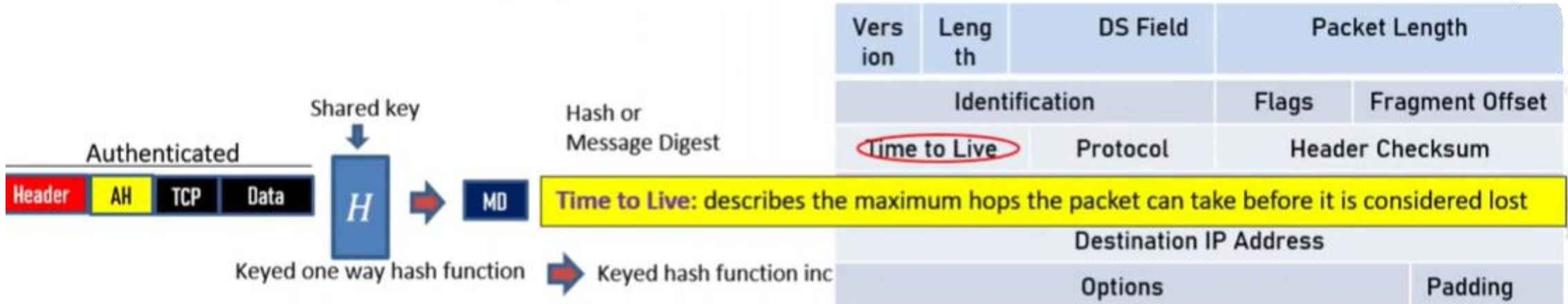
Some header fields (e.g. Time To Live (TTL)) are not used in calculation of MD



Authentication Header (AH)

Transport Mode: No protection of IP header

Some header fields (e.g. Time To Live (TTL)) are not used in calculation of MD

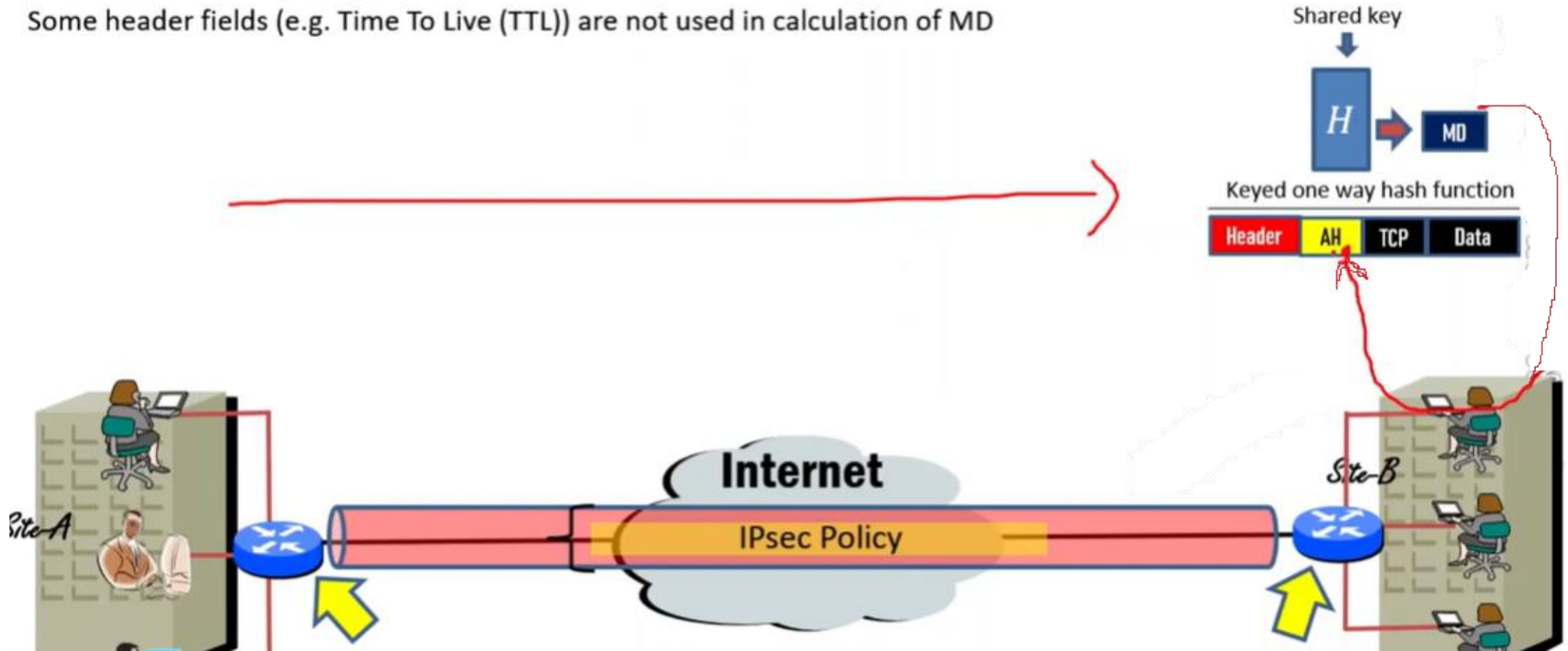


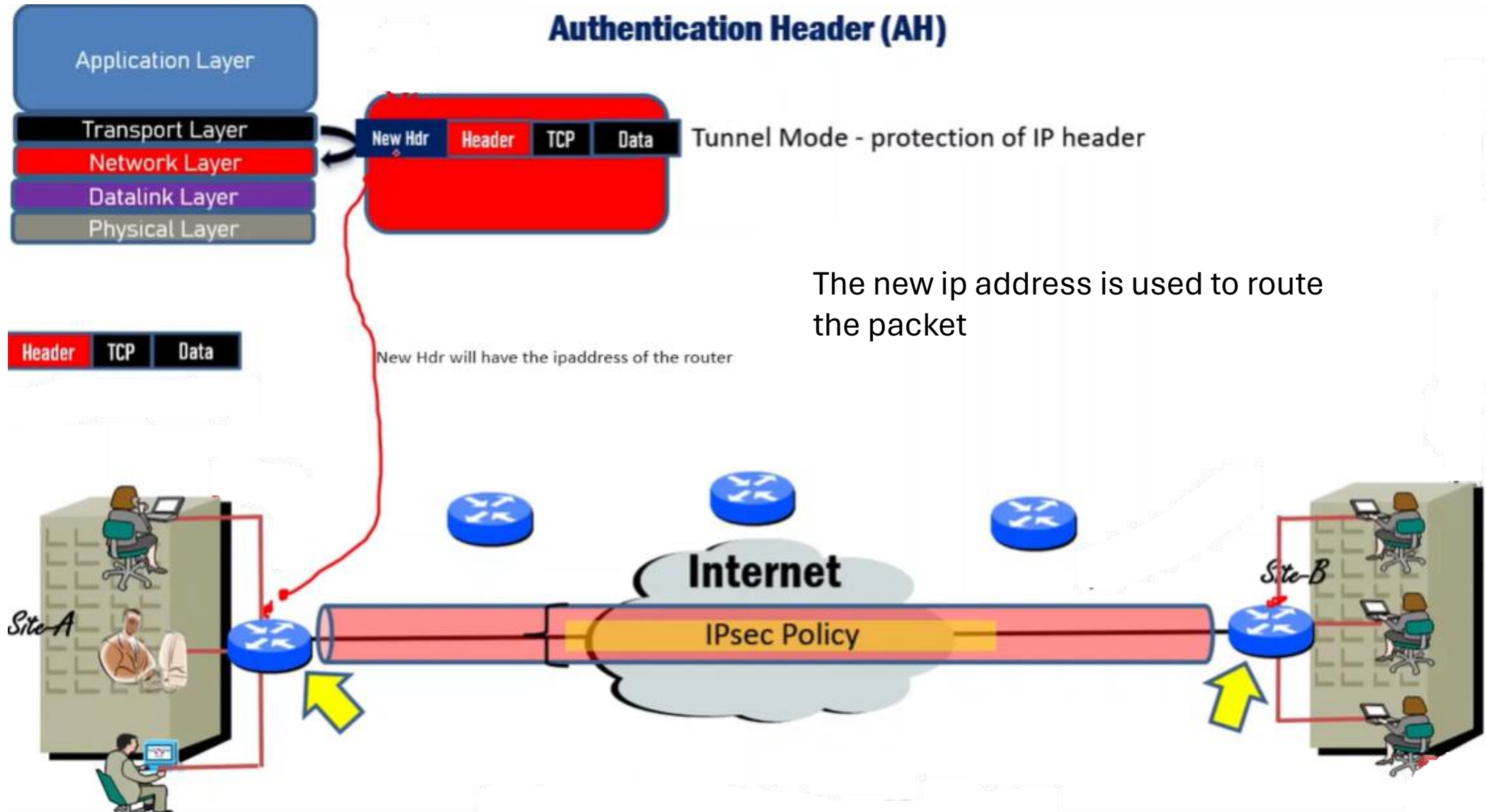
- At the destination side it will generate the message digest and compares the MD. If it matches, then the data will be accepted otherwise the data will be rejected.

Authentication Header (AH)

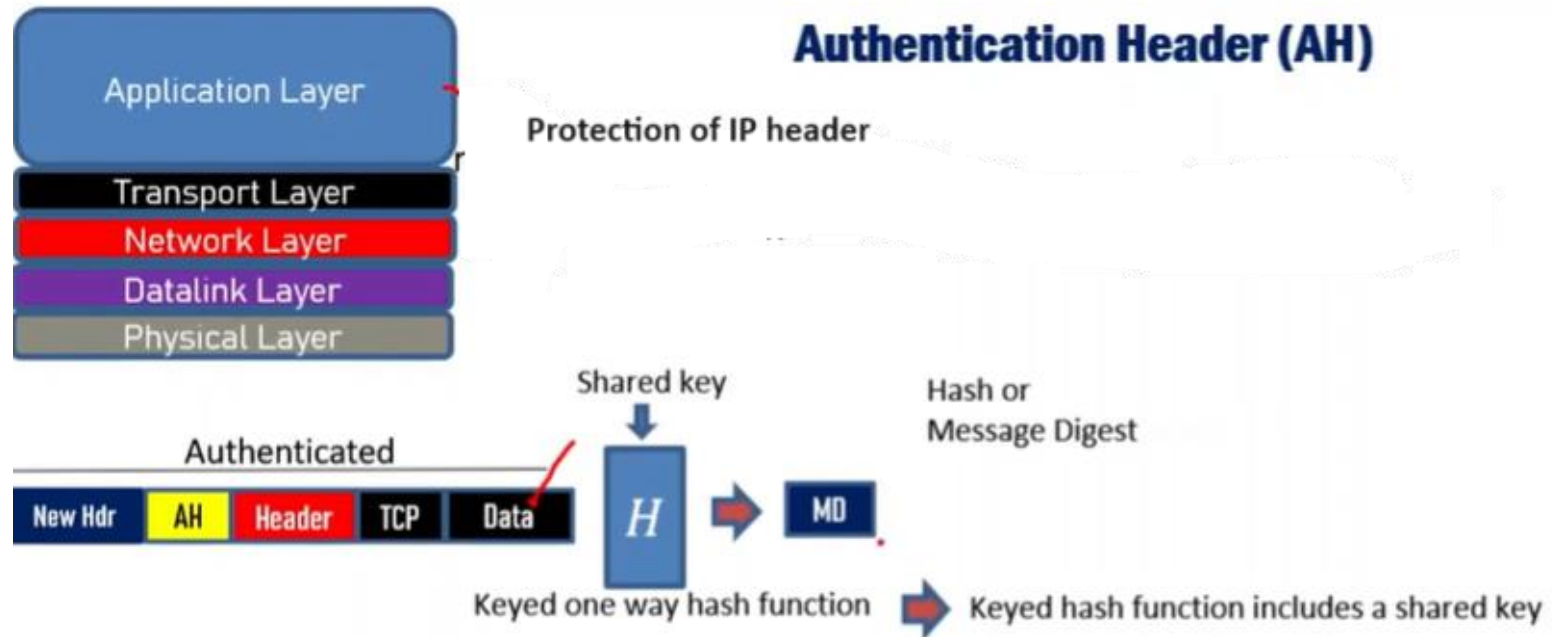
Transport Mode: No protection of IP header

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Tunnel Mode



Authentication Header (AH)

Tunnel Mode: Protection of IP header

Some header fields (e.g. Time To Live (TTL)) are not used in calculation of MD

